Experimental Insights on the effects of Explainable AI features on User satisfaction, trust, and accuracy of recommendations in Recommender System.

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Abstract: Artificial Intelligence is a critical component of recommender systems. However, this technology has resulted in some concerns, such as user dissatisfaction and distrust in the accuracy of recommendations from these systems. Our research seeks to investigate the influence of Explainable AI on user satisfaction, trust, and the accuracy of algorithmic recommendations in recommender systems. We enhance a recommender system with Explainable AI features and then perform experiments to compare a traditional recommender system with the Explainable AI-enhanced system. Through our experiment, we aim to assess the perceived transparency of recommendation algorithms in traditional recommender systems and compare them with XAI-enhanced systems features. Through our findings, we will deliver an Empirical Contribution to the research body; we will give insights into guidelines for creating more satisfactory, trustworthy and human-centered recommender systems, which will be achieved by illuminating the relationship between explainability, user satisfaction, and recommendation reliability.

CCS CONCEPTS • Xplainable AI • Recommender Systems • Human-centered systems

Additional Keywords and Phrases: Machine Learning, Artificial Intelligence.

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1 INTRODUCTION

Artificial Intelligence has become ubiquitous today and is interwoven in our everyday lives. Researchers in the research community and industry develop machine learning algorithms and leverage these algorithms to perform tasks and make decisions. This shows the importance of these algorithms in performing tasks. Therefore, machine learning algorithms must be explainable to build trust and user satisfaction in recommender systems that depend on these algorithms, of particular interest [1]. The integration of Artificial Intelligence (AI) in recommender systems, which provide users with customized content tailored to their preferences and derived from complex algorithms, has revolutionized the content delivery industry. However, these algorithms are complex and multivariate, affecting user trust and satisfaction with these

systems. To address this concern, our paper explores the potential of Explainable AI (XAI) features in addressing them. Explainable AI has the potential to enhance user experience and promote trust in Artificial Intelligence systems with AI explanations, as it offers a more human-centered approach that connects the user to the decision-making process of the recommender system. Explainable artificial intelligence provides the system's user with an understanding of the operation of the machine learning model and how it makes its decisions [2]. The explanations justify the decision or action by the machine learning algorithm. Researchers have actively researched the concept of Explainability in Artificial Intelligence due to the need to communicate trust to users in decision-making and the reason behind the results in systems that utilize machine learning algorithms, such as recommender systems [3]. Confalonieri et al. [3] wrote about the necessity for an explanation to be provided to users and regulators when decisions are taken or suggested by automated systems. This is due to more than human rights and technology acceptance. It makes a system more robust, and increasing trust is also necessary for users to know how and why decisions are made [3]. Thus, equipping an intelligent system with explainability features has become a desirable functionality in recommender systems [3], [4]. Doran et. Al [4] characterizes research on explainable AI into opaque, interpretable, and comprehensible systems. In an opaque system, the mechanism that links the user's input to the output is not visible to the user. This can arise when the Artificial Intelligence system is not open-sourced. The user can see, study, and understand how the machine learning model mathematically links their inputs to the output in interpretable systems. In this system, the user must have a high-level understanding of the system's technicalities. Comprehensible systems produce text or visual explanations with their output, which allow the user to relate the results of their outputs with their inputs. The increasing dependence of users on recommendation algorithms for consuming a wide variety of content highlights the importance of understanding the factors influencing user satisfaction and trust. Existing literature recognizes the importance of Explainable Artificial Intelligence; however, it does not deliver empirical contributions on the effects of particular XAI features on user perceptions. Our study addresses this by conducting experiments that measure how XAI features affect user satisfaction, trust, and recommendation accuracy. We will provide insights into the effectiveness of XAI in recommender systems using a combination of quantitative metrics and qualitative insights. Through a thorough systematic review of existing literature and examination of our results, we will provide an empirical contribution to the scientific research community on optimizing recommender systems to improve human-centeredness and build confidence in the decision-making of these recommendation algorithms. Our study will also illuminate the advantages of XAI-enhanced recommender systems over traditional systems and discuss the limitations and how to balance recommendation accuracy, user satisfaction, and transparency.

2 RELATED WORKS

Researchers have examined Explainable AI features and the relationship between algorithmic transparency and user experience in recommender systems [5]–[8]. The growing use of intelligent machines has resulted in a greater need for interpretable models to help humans understand them. Various XAI approaches and techniques are used based on the data type we are dealing with[1]. Gedikli et al. [9] examined user experience with recommendation systems by analyzing the effects of 10 different explanation types on users. They assessed the various explanation types and utilized qualitative semi-structured interviews to interpret and validate the results of their study. They then proposed guidelines for developing explanation interfaces for recommender systems that support the user in the decision-making process. Ali et al. [10] highlighted a four-axis framework to examine an ML model for explainability. They discussed different explainability assessment methods and proposed a novel methodological workflow for determining the criteria for the explainability of ML models. Nava Tintarev [11] conducted a study on the explanations of recommendations for a movie recommender system. They identified seven different aims of explanations and how explanations can be optimized for some of these

aims. They considered both the contents of the explanations as well as the presentation. They also developed a prototype system to test their hypothesis. Bostandjiev et al. [12] developed an interactive hybrid recommendation system that generates item predictions from multiple web sources. The system employs hybrid techniques from traditional recommender system literature and a novel interactive interface that explains the recommendation process and elicits preferences from the end user. They then compared different interactive and non-interactive hybrid techniques for computing recommendations across different APIs. Their findings indicated that explanations and interaction with a visual representation of the hybrid system increased user satisfaction and relevance of predicted content Bostandjiev et al. [12].

2.1 Textual vs. Visual Explanations

Dwivedi et al. [13] studied the benefits and preferences of explanations for personalized recommendations. They used different algorithm styles explanations and manipulated the presentation format (e.g., textual vs. visual). They show the user one textual explanation and three kinds of visual charts. They found out that most users prefer textual explanations to visualization. Initially, researchers usually tend to measure which XAI methods users prefer. Visual AI explanations can be frustrating, and user levels of expertise might affect that. Accordingly, researchers combine visual and textual elements Kouki et al. [14], and the results show significant improvement in understandability compared to only visual explanations.

2.2 UX Design for Recommender System

Platform design in recommender systems affects users' gain of the objectives of XAI, like trust and understandability. This paper aims to study the effect of the explanation in two approaches to selecting a movie [5]. One design does not describe features, while the other describes all features. They focus on the explanation's content and its presentation. Some features like reviews, cast, dialogue. They find that users like to have fewer features with more textual explanations, and they like to have more Personalized platforms and important feature information by default, with the complexity of AI-based algorithms and user demand for more control and personalized systems. The study explores using interactive explanations to enhance sense-making and user-system trust. One of the explanation types is based on Content-Based Recommendations, in which the recommendations are generated based on the genre, story keywords, and actors because the initial study found that these factors were the most significant in determining user ratings. The prototype for it is called (genre-swapping). The interactive interface allows users to explore different genre categories and observe how their choices impact the recommendation list. Users can add, delete, or combine up to three genres to create personalized recommendations that suit their preferences. This feature allows users to fine-tune their recommendations and demonstrates the explainable AI (XAI) aspect by illustrating how selecting different genres influences their viewing suggestions. They use textual descriptions accompanied by the interactive genre-swapping feature to assist users in understanding and making sense of their choices. The results indicated that interactive explanations effectively help users comprehend recommender systems and their recommendations.

3 METHODOLOGY PROPOSED

Our Research Method will be an Experimental, Between-Subjects design, and we will experiment to understand how Explainable AI features affect User satisfaction, trust, and accuracy of recommendations in recommender systems. Our Independent Variables will be the Explainable features of an XAI-enhanced recommender system, UX design features and User Experience with recommender systems. Our Dependent Variables will be User satisfaction, trust, and accuracy of recommendations.

3.1 Research Question

Our Research question is titled “How do Explainable AI features affect User satisfaction, trust and accuracy of recommendations in recommender systems?”. Our proposal aims to experiment and gain insights into the effects of explainable AI on user satisfaction, trust, and the quality of the recommendations from a recommender system.

3.2 Systematic Literature Review

We conducted a detailed and comprehensive systematic review to deliver a meticulous summary of available research that would aid in answering our research question on how Explainable AI features affect User satisfaction, trust and accuracy of recommendations in recommender systems. We searched three databases for existing literature on the topic: the IEEE, Google Scholar (GS), and ACM databases. Our search criteria included keywords that were relevant to our research question, such as “Explainable Artificial Intelligence,” “Recommender System*,” “XAI,” etc. Our search identified the following records: IEEE – 19 798; ACM – 2,011,442 and GS – 1,464,100. We screened these records by juxtaposing keywords to produce more relevant results to our search. This resulted in 12,286 records for the IEEE database, 384,885 for the ACM database, and 66,341 for the GS database.

We then applied some inclusion and exclusion criteria to screen out some records that were not relevant to our search. An example of an inclusion/exclusion criteria we utilized was the year of publication. We considered the latest literature and screened out records published earlier than 2018. We also screened out articles that had been repeated in multiple databases. For our sample population, we considered participants from an English-speaking country (the U.S.A., to be specific) and included all genders in the demographics. This resulted in 608 records for the IEEE database, 117 for the ACM and 15 for the GS database.

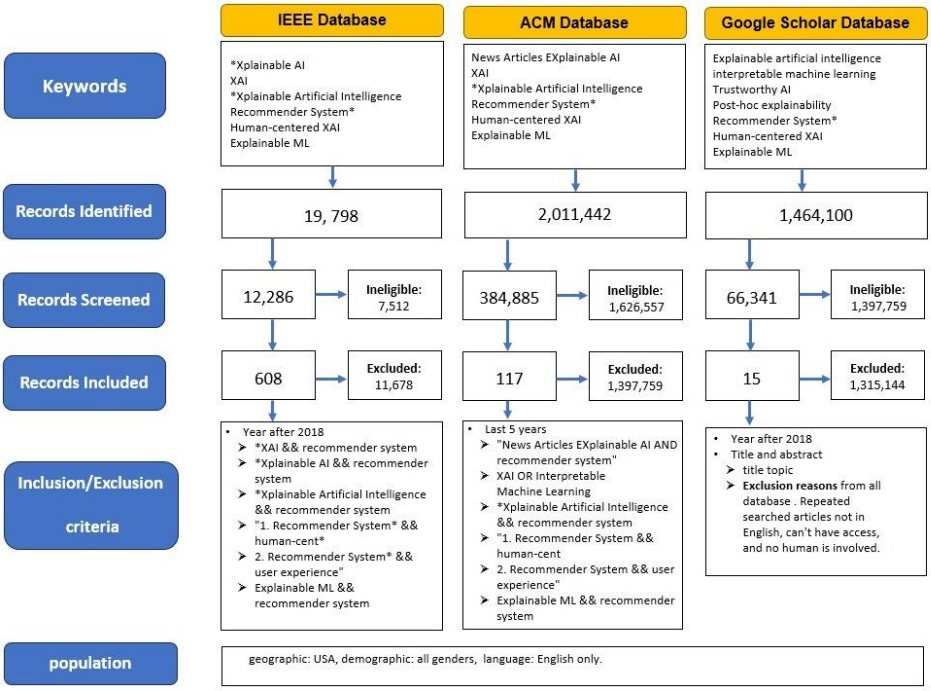


Figure 1: Systematic Literature Review

3.3 Hypothesis

h0: There is no significant difference in user satisfaction, trust and recommendation accuracy between traditional recommender systems and Explainable AI-enhanced recommender systems.

h1: There is an increase in user satisfaction and trust when explainable AI features are introduced to a recommender system than when they aren't.

3.4 Variables

Our independent variables will be UX design features and user experience with recommender systems. Our Independent Variables will have two levels or test conditions: textual explanations and visual or interactive explanations. The XAI-enhanced recommender system will provide textual explanations, and the traditional recommender system will provide visual explanations. The IV on user experience with recommender systems will have two levels or test conditions: experience with Recommender systems and inexperience with recommender systems. Our dependent Variables will be User satisfaction, trust, and accuracy of recommendations. We have two independent variables; therefore, our experimental design will have two primary and one interaction effect. We will measure the Dependent variables using questionnaires. We will utilize a System Usability Scale (SUS) survey to measure User Satisfaction and Recommendation Accuracy; we will apply a Likert scale with a rating scale of 1 – 5 to measure Trust, where one will signify a highly positive response, and five will signify a highly negative response; Three will signify a neutral response.

Independent Variables	**Effects**		**Total**
	Main Effect	**2-way Interaction Effect**	
UX Design Features	1	-	1
User Experience, UX Design Features	2	1	3

Table 1: The number of Effects increases as IVs increase.

We will introduce Control Variables to address factors that could impact the Dependent Variables. We will control the characteristics of the study participants by ensuring that the participants are above 18 years of age, are from an English-speaking country (U.S.A.), and are native English Speakers. We will also ensure the UX Design features are consistent during the experiment. Confounding variables are unmeasured or unaccounted for and can impact our experiment. We will, therefore, cater to this by considering the level of experience of the users with recommender systems. We will also consider the mood and countenance of the study participants during each experiment by conducting an icebreaker before the start of the experiment as a counterbalance to this effect. This would enhance external validity by making the experiment more generalizable and increase internal validity**.**

3.5 Experimental design

Our experimental design will consider all the factors involved in testing a hypothesis. We will define our IVs and DVs and design the experimental procedure, tasks, participants, and other factors. We will manipulate our Independent Variables, namely the UX design features and user experience with recommender systems. Will the user be more satisfied and trust the recommender system more if it is XAI enhanced?

3.6 Apparatus

We will utilize two recommender systems, Netflix and MovieLens, to manipulate our independent variables. For this experiment, Netflix will be the traditional recommender system, and MovieLens will be the XAI-enhanced recommender system. We will use questionnaires to collect data on our dependent variables, namely User satisfaction, trust, and accuracy of recommendations. The questionnaires will be close-ended, and we will apply a Likert Scale and a System Usability scale in the questionnaires.

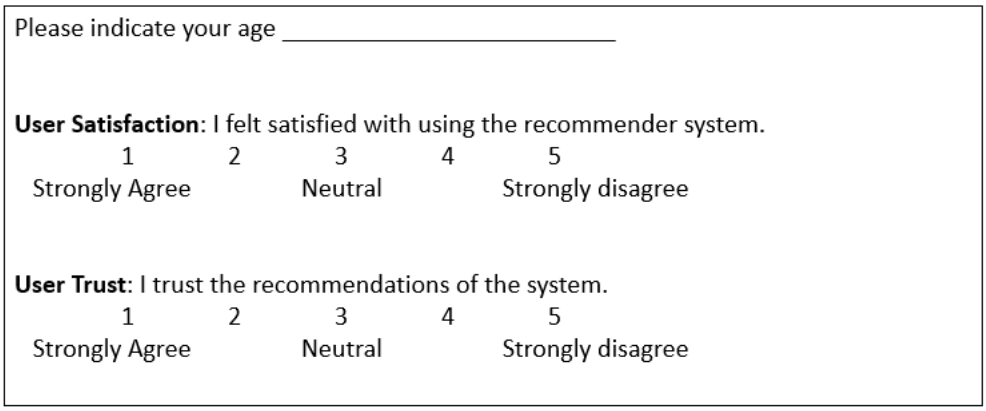


Figure 1: Questionnaire Design

3.7 Experimental Procedure

We will conduct a pre-test and a post-test. Our sample will be divided into a control group and an experimental group. Participants in the control group will engage the traditional recommender system, while the other group will engage the XAI-enhanced recommender system. The control group will serve as an alternative to test our hypothesis. We will have two conditions for our Independent Variables: Textual and Visual Explanations. The experiment will have a duration of 6 months in which participants from both groups will engage both Netflix and MovieLens once every week to get recommendations for their movies. The characteristics of both Netflix and MovieLens will be the same; only the UX design characteristics, namely the textual and visual explanations, will vary. Upon completing the six months, we will then ask participants from both groups to compare the recommendations they obtained on both platforms. The control group will not interact with the independent variables. We will also present the conditions in different orders to our different participants to promote randomization and cater to order effects. Each participant will experience one level of each independent variable. Thus, five participants will experience textual explanations for each group, while the other five will experience visual explanations. For our Experiment, we will ask both study groups to engage the recommender systems and get recommendations from the traditional and XAI recommender systems. They will get both textual explanations and visual explanations. There will also be ten observations per condition. We will then make our observations of both conditions from the ten movies. We will deploy a questionnaire with ratings to get qualitative feedback on user satisfaction and trust from both study groups. We will also use a questionnaire to measure the recommendations' accuracy. After the design of our experiment, we will obtain ethics approval through Clemson University's Institutional Review Board (IRB) [15]. We have prepared the IRB documents, which will inform the participants of the nature of the research, the research

methodology and any risks or benefits associated with the study. It will also detail their right to consent, anonymity and confidentiality.

3.8 Participants

The population of our study will be adult participants from the USA; then, we will use an IP filter to be sure they are in a specific country. The participants will have a mean age of 39.3 (range = 18 to 65 years old) and an annual household income below $ 5,000 to $ 70,000. Our sample size will be a random sample of 308 participants involved in this study in 2 groups of 154 people. One group will be the control group, whose participants have yet to gain experience with recommender systems. They will also engage the traditional recommender system Netflix. The experimental group will consist of 154 participants trained in XAI methods and will engage the XAI-enhanced recommender system MovieLens.

3.9 Data Analysis

The data on user satisfaction, trust and recommendation accuracy will be categorized based on the experimental condition (control group against experimental group). We will analyze the user feedback from the experimental and control groups. We will also measure the accuracy of the recommendations from both groups. We will use statistical analysis **(Chi-Square test)** software to compare the results between classifications.

4 RESULTS (EXPECTED)

We expect to collect 308 participants to answer the survey in a crowdsourcing platform; this number is higher than the calculated with G*power () because it is necessary to review for outliers. The sample could include 55% females and 45% males with an average of 39.3 (range = 18 to 65 years old and standard deviation of 15 years). We expected participants in crowdsourcing to come from the USA. A requirement for participation is to have a Netflix account. Most of them (89%) reported using Netflix at weekends, while the other 11% use it all week. We found eight visual and 13 textual features based on the Netflix or Movielens webpages' features. The feature awareness will be classified according to the number of features used; we will be ranging them from "Experts" (uses all features), "moderate experts" (uses more than 17 features), "Some expertise" (14 features), "Mostly Novice" (uses10 features) "Near Novice" (use more than five features) "Novice" (use less than five features). Figure 2 shows the percentage of this classification.

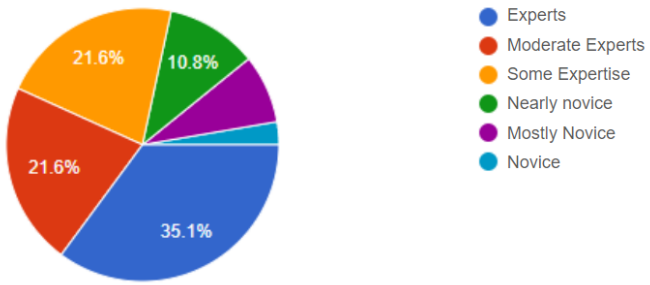


Figure 2: Percentage Classification of Experience with Recommender System

We expected to do another classification according to the following characteristics:

Visual maximizers. Users that only use visual features to select a movie.
Selective. Users that only ask friends or family to select a movie.
Feature Balancers. Users select almost the same features in visual and textual categories.
Textual maximizers. Users that only use textual features to select a movie.
Privacy concern. These users use the features to choose the movie but are afraid of clicking them because of the privacy of their profiles.
Time minimalist. These users view the features for less than 1 minute and choose a movie.
To make this classification, we will ask about the use of visual and textual features, which can be seen in Figures 4 and 5

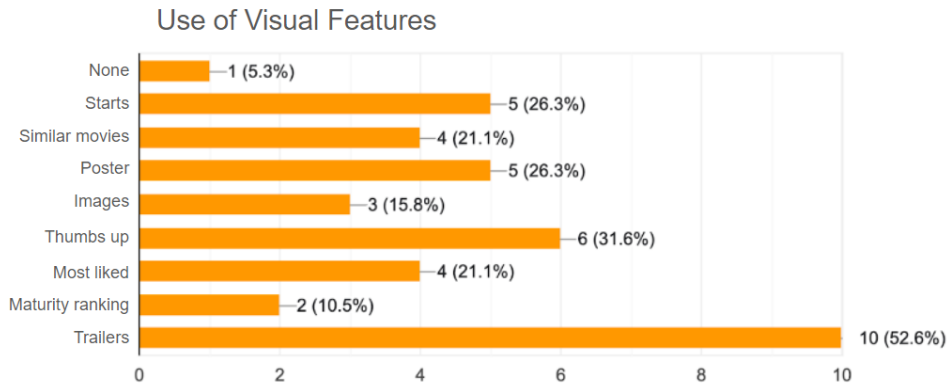


Figure 3: Use of Visual features

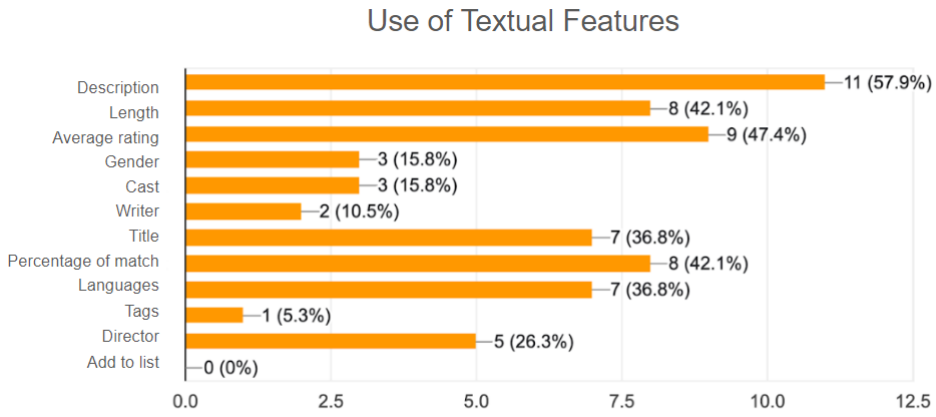


Figure 4: Use of Textual Features

We expect to use both classifications to build a Shaded mosaic plot to analyze the interaction of the classification with a Chi-square test (Figure 6)

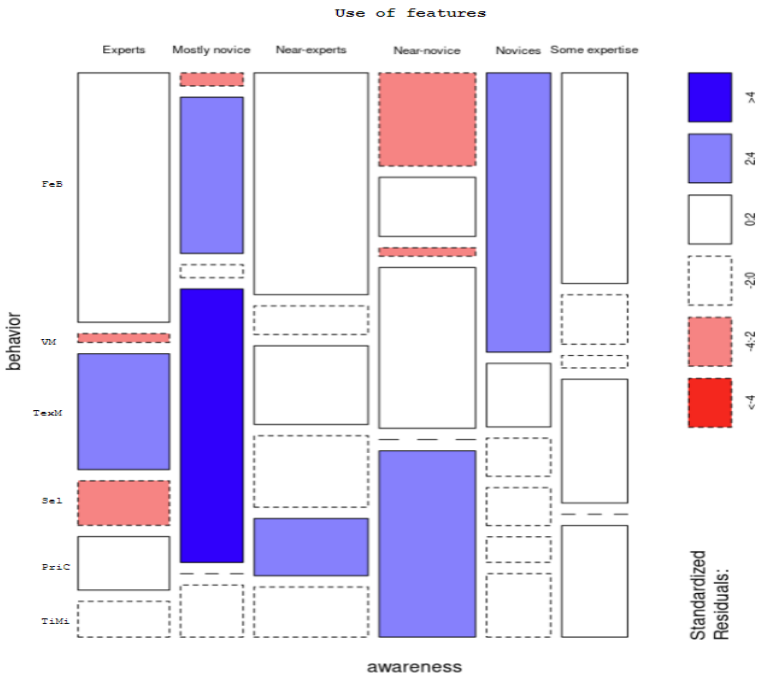


Figure 5: Mosaic Plot

There was a significant association between feature awareness and feature use behavior.

Underrepresented: Visual Maximizer-Mostly novice (-1.436), Visual Maximizer- Near novice (-2.004), Minimalists-Expert (-2.138), Experts-Minimalists (-2.138), Experts-Textual Maximizer (-2.067).

Overrepresented: Privacy Concern-Mostly novice (4.2222), Visual Maximizer-Experts, Experts-Visual Maximizer (3.161). Upon analyzing the effect, we will find that the odds of visual maximizer rather than privacy concern were 9.6353 times higher for experts than all other feature awareness levels.

chi^2 = 135.603 d.f. = 25 p = 3.601029e-17

5 DISCUSSIONS

Overall, the mosaic plot suggests that participants in the "Textual Features" condition understand less how the recommendations are made than the other conditions, as indicated by the giant box in the low understandability category. However, there appears to be a weak association between the two variables based on the need for consistent cell shading patterns. With this study, we will find which characteristics are more useful for the understandability of users about recommended systems. We will experiment with the survey that gives us the insights to classify the users and then conduct a chi-square analysis to examine the relationship between the textual/visual explanations and User satisfaction, trust, and accuracy of recommendations. These findings suggest that visual features are more important for understandability or that textual features are more critical than visual ones. Also, it may be a balanced combination of features. We expect to observe associations in the use of features to understand better how the algorithms must be changed to give better recommendations and better explanations to the user.

6 LIMITATIONS

Researchers' future vision for XAI is to have recommendations for algorithm and UI design work together. Yet, there needs to be more research regarding the correlation between UX design for recommendation systems and the impact of each design element. Assuming users get familiar with the categories and descriptions in the system and understand the algorithm behind the recommender rows and ranking, many movie streaming platforms employ personalized algorithms that continually adapt, resulting in frequent interface modifications. Nevertheless, an unstable interface may consistently trigger user frustration and recurring confusion.

The demand for XAI is burgeoning, especially for platforms involved in sensitive decision-making like health and banking. However, in entertainment platforms like video streaming, the same level of XAI may be optional. Users may prioritize the straightforward enjoyment of content without requiring in-depth explanations. A study was made by Tran, T. N. T., et al. [16] to study whether users appreciate explanations on movie-based recommender systems platforms, and they found that users only sometimes appreciate explanations! They only like to see explanations if they are unsatisfied with the system.

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Contribution Statement

We organized the team with a WhatsApp group chat; this led us to do it asynchronously. Everyone participated in brainstorming in the different areas of the document, reading articles, and looking for the best approaches for the experiment. The report's components were agreed upon and divided equitably among team members. The sections were distributed as the following: Raphael: Abstract, Introduction, Methodology and some parts of the Literature Review (Consulted ChatGPT with Prompt "Please give me tips for designing an experiment on explainable AI and recommendation systems"); Laila: Related Works, Limitations, IRB application, advertising messages and exempt-adult-consent; Cecilia: Results (following an exercise of class Measurements and Evaluations), Discussion and Contributorship statement, questionnaire for planning experiment and results.

We added to the paper's overall coherence and flow through editing and rewriting passages. At the final stage of the task, all participants comprehended and approved our completed version.